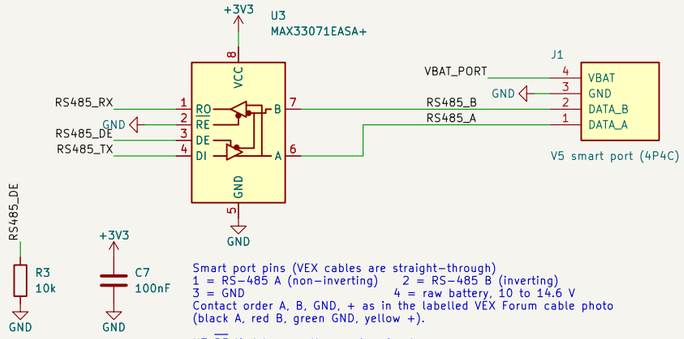
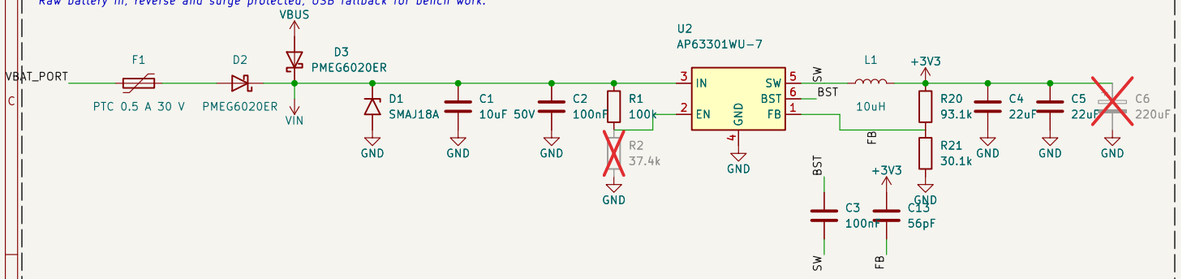
V5 smart port and RS-485

Brain to board over one smart port. Half duplex, so the board can also send back.



Power input and 3.3 V buck

Raw battery in, reverse and surge protected, USB fallback for bench work.



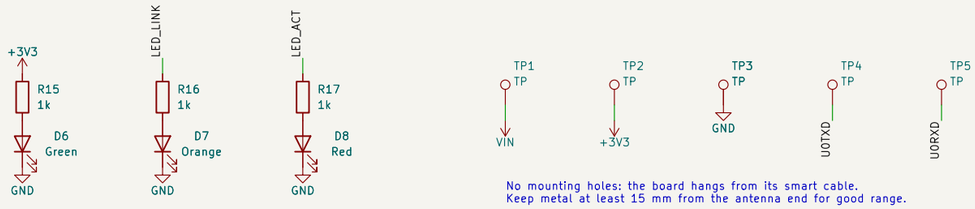
U2 AP63301: adjustable buck, 0.8 V FB, 500 kHz forced PWM, 3 A rated, 0.57 A worst case here.
Vout = 0.8 V x (1 + 93.1k / 30.1k) = 3.27 V. C13 56 pF feed-forward and C4, C5 2 x 22 uF per datasheet.
L1 10 uH (datasheet range 2.2 to 10 uH): ripple 0.51 A at 14.6 V in. Isat 4.6 A at 20% drop.
C6 220 uF is DNP: the datasheet characterises 22 to 68 uF out. Fit only if a load-step test shows a need.
Port current worst case: about 220 mA peak, 50 mA average at 10 V battery.
D2 blocks reverse polarity. D3 lets USB run the board on the bench, and D2 stops it back-feeding the brain.
D1 18 V standoff clears a 14.6 V battery and clamps under the 35 V buck limit.
R1 enables the buck from VIN. R2 (DNP) adds a UVLO divider, but fitted it stops USB start-up.



ERC power flags (no parts)

Status LEDs and test points

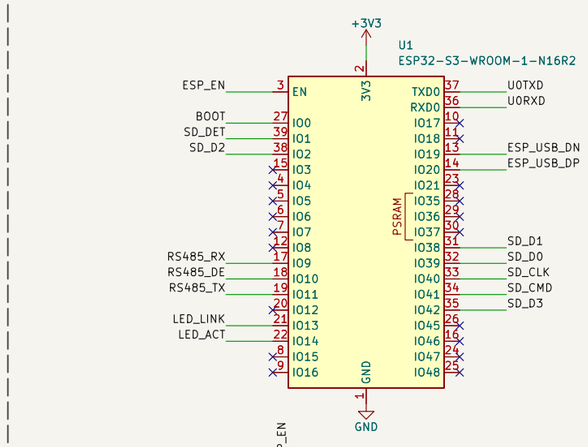
About 1.3 mA per LED through 1 k. Blue and InGaN green need too much voltage for 3.3 V.



No mounting holes: the board hangs from its smart cable.
Keep metal at least 15 mm from the antenna end for good range.

ESP32-S3 module

WiFi 2.4 GHz and Bluetooth LE 5. Native USB, no bridge chip.



SW1 RESET, SW2 BOOT. Hold BOOT, tap RESET for the ROM bootloader.

Firmware pin map

All signals 3.3 V. GPIO numbers are ESP32-S3 IO numbers.

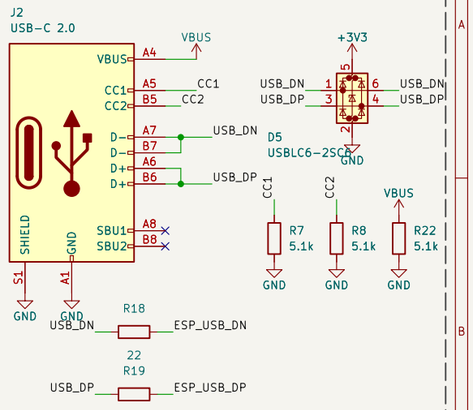
RS-485 RX IO9 DE IO10 TX IO11 (UART_MODE_RS485_HALF_DUPLEX, RTS = IO10)
USB D- IO19 D+ IO20 (USB Serial/JTAG, auto-reset into bootloader)
microSD CLK IO40 CMD IO41 D0 IO39 D1 IO38 D2 IO2 D3 IO42 DET IO1 (low = card in)
LEDs LINK IO13 ACT IO14 (high = on)
Buttons RESET = EN, BOOT = IO0
UART0 TX IO43 (TP4), RX IO44 (TP5): ROM boot log

Brain side: vexGenericSerialEnable(port, 0) then
vexGenericSerialBaudrate(port, 230400). PROS: pros::Serial.

Module N16R2 has quad PSRAM: set PSRAM mode to Quad, not Octal.
Strapping pins IO3, IO45, IO46 are left floating on purpose
(internal defaults give SPI boot and 3.3 V flash).

USB-C

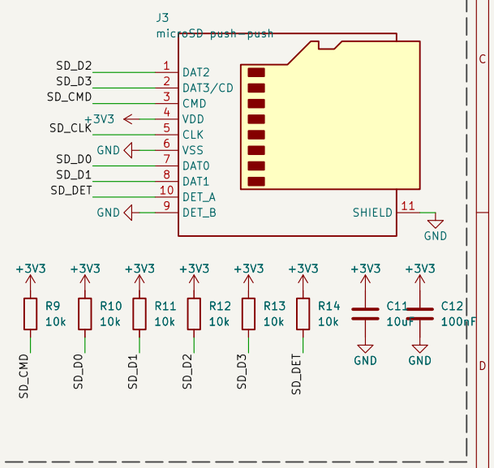
Flashing, JTAG, wired console and bench power.



5.1 k CC pulldowns: USB sink, 5 V only.
R18/R19 22 ohm series close to U1. D+/D- as a 90 ohm pair.
R22 drains D3 leakage so VBUS reads 0 V on port power and a host will attach.

microSD logging

Keeps logging when the wireless link drops.



Generated design. Verify the smart port pinout with a meter before first plug-in.
Single smart port, RS-485 half duplex, powered from the V5 battery
stellaNova: ESP32-S3 WiFi/BLE debugger for the VEX V5 smart port

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